

1 Gain law

(bars denote normalized quantities: lengths by R ; $a_o := \Delta t / (\gamma_N R)$ [1/pN] and $a_{\text{nom}} := \Delta t / \gamma_N = a_o R$ [$\mu\text{m}/\text{pN}$]; $\bar{a} := a / a_{\text{nom}} = a / (a_o R) = 1/c$ for the physical gain a [$\mu\text{m}/\text{pN}$]; $\bar{f} := a_o f = f / f_R$ with $f_R := \gamma_N R / \Delta t = 1/a_o$ [pN], so that $\bar{a} \bar{f} = \Delta \bar{w}$; $0 \leq \bar{a} \leq 1$ on $\bar{w} - \bar{w}_s \geq 1/b$; hats sit on the normalized quantities, $\hat{a}_w, \hat{b}, \delta \hat{w}; e_\theta = \theta - \hat{\theta}$)

$$\bar{a}_{\text{true}}(\bar{w}) = 1 - \frac{9}{8\bar{w}} + \frac{1}{2\bar{w}^3} - \frac{57}{100\bar{w}^4} + \frac{1}{5\bar{w}^5} + \frac{7}{200\bar{w}^{11}} - \frac{1}{25\bar{w}^{12}}$$

$$\bar{a}'_{\text{true}}(\bar{w}) > 0 \quad \text{on} \quad \bar{w} > 1 \quad \implies \quad \bar{a}' = b_{\text{true}}(\bar{a})(1 - \bar{a})^2 \quad \text{exactly,} \quad b_{\text{true}} := \frac{\bar{a}'_{\text{true}}}{(1 - \bar{a}_{\text{true}})^2}$$

$$\begin{aligned} \bar{a}_{\text{true}}(1) = 0, \quad \bar{a}'_{\text{true}}(1) = 1 &\implies b_{\text{true}}(\bar{a} = 0) = 1 \\ 1 - \bar{a}_{\text{true}} = \frac{9}{8\bar{w}} + \mathcal{O}(\bar{w}^{-3}) &\implies b_{\text{true}}(\bar{a} \rightarrow 1) = \frac{8}{9} \end{aligned}$$

$$\begin{aligned} \bar{a}'(\bar{a}, b) &:= b(1 - \bar{a})^2 \\ a_w &= a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R} \\ \bar{f} &= a_o f = \frac{f}{f_R}, \quad f_R = \frac{\gamma_N R}{\Delta t} = \frac{1}{a_o} \end{aligned}$$

2 Jacobians

$$\begin{aligned} \frac{\partial \bar{a}'}{\partial \bar{a}} &= -2b(1 - \bar{a}) \\ \frac{\partial \bar{a}'}{\partial b} &= (1 - \bar{a})^2 = \frac{\bar{a}'}{b} \end{aligned}$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \\
\delta \bar{w}_1[k+1] &= \delta \bar{w}_2[k] \\
\delta \bar{w}_2[k+1] &= \delta \bar{w}_3[k] \\
\delta \bar{w}_3[k+1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - (1 - \lambda_c)(m_1[k] + m_2[k]) - w_T[k] + (1 - \lambda_c) n_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] \right. \\
&\quad \left. + (1 - \lambda_c)(m_1[k] + m_2[k]) + w_T[k] - (1 - \lambda_c) n_w[k] \right] \\
b[k+1] &= b[k] \\
m_1[k+1] &= w_T[k] \\
m_2[k+1] &= m_1[k]
\end{aligned}$$

$$\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k-i] \\
w_T[k] &= \bar{a}_w[k] \bar{f}_T[k] \\
\varepsilon_w[k] &= w_T[k] + (1 - \lambda_c)(w_T[k-1] + w_T[k-2]) - (1 - \lambda_c) n_w[k] \\
&= w_T[k] + (1 - \lambda_c)(m_1[k] + m_2[k]) - (1 - \lambda_c) n_w[k]
\end{aligned}$$

4 Process model

$$\mathbf{x} = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, b, m_1, m_2]^\top \quad (d=2; m_1 = w_T[k-1], m_2 = w_T[k-2])$$

5 Estimator

$$\begin{aligned}
\delta \hat{w}_1[k+1] &= \delta \hat{w}_2[k] \\
\delta \hat{w}_2[k+1] &= \delta \hat{w}_3[k] \\
\delta \hat{w}_3[k+1] &= \lambda_c \delta \hat{w}_3[k] - (1 - \lambda_c)(\hat{m}_1[k] + \hat{m}_2[k]) \\
\hat{a}_w[k+1] &= \hat{a}_w[k] + \hat{b}[k] (1 - \hat{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] + (1 - \lambda_c)(\hat{m}_1[k] + \hat{m}_2[k]) \right] \\
\hat{b}[k+1] &= \hat{b}[k] \\
\hat{m}_1[k+1] &= 0 \\
\hat{m}_2[k+1] &= \hat{m}_1[k]
\end{aligned}$$

6 Error dynamics

(True – Estimator; coefficients at the estimate)

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c e_{\delta\bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - (1 - \lambda_c)(e_{m_1}[k] + e_{m_2}[k]) - w_T[k] + (1 - \lambda_c) n_w[k] \\
e_{\bar{a}_w}[k+1] &= \left[1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} - 2\hat{b}(1 - \hat{a}_w)(\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3 + (1 - \lambda_c)(\hat{m}_1 + \hat{m}_2)) \right] e_{\bar{a}_w}[k] \\
&\quad + (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 (e_{\delta\bar{w}_3}[k] + e_{m_1}[k] + e_{m_2}[k]) \\
&\quad + (1 - \hat{a}_w)^2 (\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3 + (1 - \lambda_c)(\hat{m}_1 + \hat{m}_2)) e_b[k] \\
&\quad + \hat{b}(1 - \hat{a}_w)^2 (w_T[k] - (1 - \lambda_c) n_w[k]) \\
e_b[k+1] &= e_b[k] \\
e_{m_1}[k+1] &= w_T[k] \\
e_{m_2}[k+1] &= e_{m_1}[k]
\end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 & -(1 - \lambda_c) & -(1 - \lambda_c) \\ 0 & 0 & (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 & 1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} & (1 - \hat{a}_w)^2 & \times (\Delta\bar{w}_d + (1 - \lambda_c) \delta\hat{w}_3 + (1 - \lambda_c)(\hat{m}_1 + \hat{m}_2)) & (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{q}[k] = w_T[k] \begin{bmatrix} 0, 0, -1, \hat{b}(1 - \hat{a}_w)^2, 0, 1, 0 \end{bmatrix}^\top - (1 - \lambda_c) n_w[k] \begin{bmatrix} 0, 0, -1, \hat{b}(1 - \hat{a}_w)^2, 0, 0, 0 \end{bmatrix}^\top$$

7 Measurement and innovation

$$\begin{aligned}
\nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k - d] \\
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k - d] + n_a[k]
\end{aligned}$$

$$\bar{a}_w[k - d] = \bar{a}_w[k] - b(1 - \bar{a}_w[k])^2 \nabla_d \bar{w}_d[k]$$

$$\begin{aligned}
\frac{\partial y_2}{\partial \bar{a}_w} &= 1 + 2\hat{b}(1 - \hat{a}_w) \nabla_d \bar{w}_d \\
\frac{\partial y_2}{\partial \hat{b}} &= -(1 - \hat{a}_w)^2 \nabla_d \bar{w}_d
\end{aligned}$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + 2\hat{b}(1 - \hat{a}_w)\nabla_d \bar{w}_d & -(1 - \hat{a}_w)^2 \nabla_d \bar{w}_d & 0 & 0 & 0 \end{bmatrix}$$

$$L[k] = P^f H^\top (H P^f H^\top + R)^{-1}$$

$$e_{y_1}[k] = e_{\delta \bar{w}_1}[k] + n_w[k]$$

$$e_{y_2}[k] = \left[1 + 2\hat{b}(1 - \hat{a}_w)\nabla_d \bar{w}_d \right] e_{\bar{a}_w}[k] - (1 - \hat{a}_w)^2 \nabla_d \bar{w}_d e_b[k] + n_a[k]$$

8 Process and measurement noise

$$\gamma(\bar{w}) = \gamma_{Nc}(\bar{w})$$

$$\sigma_{\bar{f}_T}^2 = a_o^2 \frac{4k_B T \gamma(\bar{w})}{\Delta t}$$

$$\kappa_T := \frac{4k_B T a_o}{R} = \frac{4k_B T a_{\text{nom}}}{R^2}$$

$$\sigma_{w_T}^2[k] = \text{Var}(\bar{a}_w[k] \bar{f}_T[k]) = \kappa_T \bar{a}_w[k] \rightarrow \kappa_T \hat{a}_w[k]$$

$$\sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2}$$

$$\mathbf{g}_T = [0, 0, -1, \hat{b}(1 - \hat{a}_w)^2, 0, 1, 0]^\top$$

$$\mathbf{g}_n = (1 - \lambda_c)[0, 0, -1, \hat{b}(1 - \hat{a}_w)^2, 0, 0, 0]^\top$$

$$Q[k] = \kappa_T \hat{a}_w[k] \mathbf{g}_T \mathbf{g}_T^\top + \sigma_{n_w}^2 \mathbf{g}_n \mathbf{g}_n^\top$$

$$Q_{33} = \kappa_T \hat{a}_w + (1 - \lambda_c)^2 \sigma_{n_w}^2$$

$$Q_{34} = Q_{43} = -\hat{b}(1 - \hat{a}_w)^2 Q_{33}$$

$$Q_{44} = \hat{b}^2(1 - \hat{a}_w)^4 Q_{33}$$

$$Q_{36} = Q_{63} = -\kappa_T \hat{a}_w$$

$$Q_{46} = Q_{64} = \hat{b}(1 - \hat{a}_w)^2 \kappa_T \hat{a}_w$$

$$Q_{66} = \kappa_T \hat{a}_w$$

$$Q_{55} = 0$$

(white-container arm, `ma2_aug = false`: no m_1, m_2 , and the whole $\text{Var}(\varepsilon_w)$ sits in Q_{33})

$$\text{Var}(\varepsilon_w)[k] = \kappa_T \left(\hat{a}_w[k] + (1 - \lambda_c)^2 (\hat{a}_w[k-1] + \hat{a}_w[k-2]) \right) + (1 - \lambda_c)^2 \sigma_{n_w}^2$$

$$\begin{aligned}
R &= \text{diag}(R_1, R_2) \\
R_1 &= \sigma_{n_w}^2 \\
R_2 &= K_{\text{var}} I F_{\text{var}} (\hat{a}_w + \xi)^2 + d Q_{44} \\
K_{\text{var}} &= \frac{2a_{\text{cov}}}{2 - a_{\text{cov}}} \\
\sigma_{\delta \bar{w}_r}^2 &= C_{dpmr} \kappa_T (\hat{a}_w + \xi) \\
\xi &= \frac{C_n \sigma_{n_w}^2}{C_{dpmr} \kappa_T}
\end{aligned}$$

Seed and prior

$$\begin{aligned}
\bar{a}(\bar{w}) &= 1 - \frac{1}{b(\bar{w} - \bar{w}_s)} \\
\left. \frac{\partial \bar{a}}{\partial b} \right|_{\bar{w}} &= \frac{1}{b^2(\bar{w} - \bar{w}_s)} = \frac{1 - \bar{a}}{b} \\
\left. \frac{\partial \bar{a}}{\partial \bar{w}_s} \right|_{\bar{w}} &= -\bar{a}'
\end{aligned}$$

$$\begin{aligned}
\hat{b}[0] &= b_{\text{true}}(\bar{a} \rightarrow 1) = \frac{8}{9} \\
P_{55}[0] &= P_{bb} = \left[\sup_{\bar{w} \in \mathcal{E}} |b_{\text{true}}(\bar{w}) - \frac{8}{9}| \right]^2 \\
\hat{a}_w[0] &= 1 - \frac{1}{\hat{b}[0] (\bar{w}[0] - \hat{w}_s)}
\end{aligned}$$

$$\begin{aligned}
P_{44}[0] &= (\hat{a}'[0])^2 P_{\bar{w}_s} + \left(\frac{1 - \hat{a}_w[0]}{\hat{b}[0]} \right)^2 P_{bb} + \text{floor}_a^2 \\
P_{45}[0] &= \frac{1 - \hat{a}_w[0]}{\hat{b}[0]} P_{bb} \\
P_{\{1,2,3,6,7\}}[0] &= \text{DARE} \left(F_e^{\{1,2,3,6,7\}}[0], H_1^{\{1,2,3,6,7\}}, Q^{\{1,2,3,6,7\}}[0], R_1 \right)
\end{aligned}$$

$$\begin{aligned}
e_{\bar{a}}(\bar{w})|_{\bar{w}_s} &= -\bar{a}'(\bar{w}) e_{\bar{w}_s} \propto \frac{1}{b(\bar{w} - \bar{w}_s)^2} \\
e_{\bar{a}}(\bar{w})|_b &= \frac{e_b}{b^2(\bar{w} - \bar{w}_s)}
\end{aligned}$$









